

# Wildfire Smoke Prediction

# Model Architecture and Design Choice

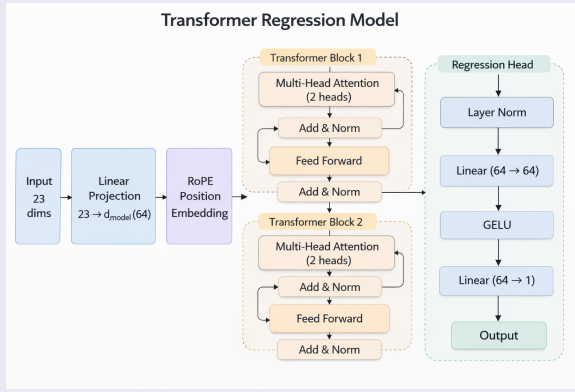
## Transformer Model Structure

### Architecture

- Transformer encoder
- Sequence input: **96 hours**
- $d_{\text{model}} = 64$
- 2 attention heads
- 2 Transformer layers
- dropout = 0.1

Total parameters size: 0.106M

Total parameters intentionally **kept small** to reduce overfitting.



## Why Not More Complex Alternatives?

We experimented with **more complex architectures** inspired by LLaMA:

- SwiGLU feed-forward layers
- Larger hidden dimensions
- Deeper Transformer blocks

However:

- Dataset size relatively small: 26245 data entries(12.05 MB) per Zone
- Models trained **separately per zone**
- Complex variants did **not improve stability or performance**

Final decision:

**Use a simpler Transformer block with strong regularization.**

## Training Setup

Optimizer: **AdamW**

Loss: **MSELoss**

Training parameters:

- Batch size: 256
- Epochs: 10
- Learning rate:  $5e-4 \rightarrow 5e-6$  (cosine decay)
- Weight decay: 0.05

Hyperparameter search was performed using **manual exploration**.

## Feature and Window Experiments

Feature set includes:

- PM2.5 lags and rolling statistics
- Fire activity indicators (FRP, fire counts)
- Temporal encoding (hour, weekday, month)

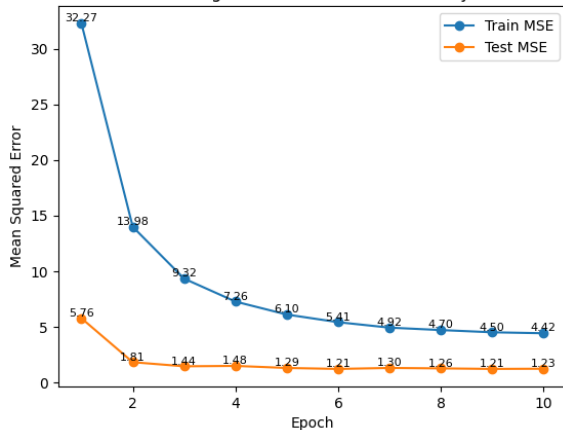
Window length experiment:

12h, 24h, 48h, 72h, **96h**, 168h

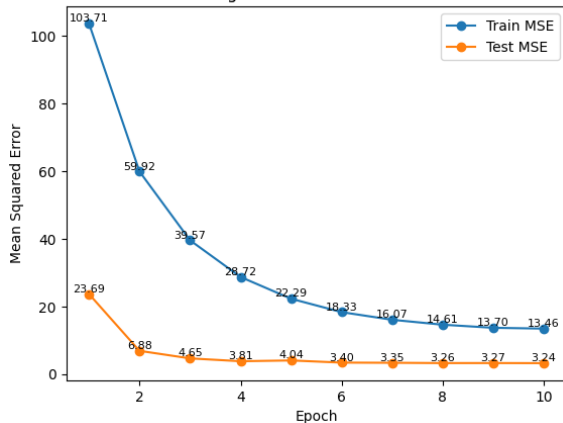
Best performance observed at **96 hours**.

# Training Dynamics

Training Curve — Lower Fraser Valley



Training Curve — Southern Interior



Training monitored using **train and test loss curves** across 10 epochs.

## Final Model Performance

| Zone                | RMSE  | MAE   | R <sup>2</sup> | N_test |
|---------------------|-------|-------|----------------|--------|
| Lower Fraser Valley | 1.108 | 0.659 | 0.843          | 8710   |
| Southern Interior   | 1.801 | 1.230 | 0.863          | 8710   |

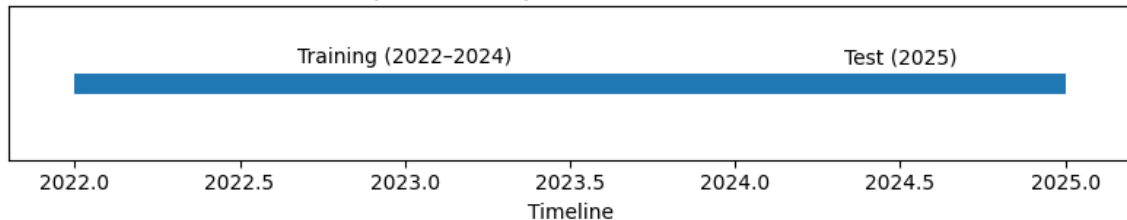
# Validation Strategy and Leakage Prevention

## Time-Series Evaluation

Random data splits are **inappropriate for time-series data**, because they allow future observations to influence training.

Instead we use **temporal splitting**.

Temporal Data Split for Model Evaluation



Train: **2022–2024**

Test: **2025**

## Leakage Prevention

Several safeguards were applied:

- StandardScaler **fit only on training data**
- Test data transformed using training statistics
- Sequence windows constructed **after dataset split**
- No window crosses train–test boundaries

## Zone-Specific Modeling

Models trained **independently** for each region:

- Lower Fraser Valley
- Southern Interior

This avoids mixing **regional wildfire patterns**.

Model predictions follow **PM2.5 trends in the unseen test year**, demonstrating generalization to new wildfire seasons.

